Carleano Chang

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Education

University of Illinois Urbana-Champaign — **Aug 2025 - May 2027**
Master of Computer Science; GPA: 3.94/4.0 — Urbana-Champaign, IL

National Tsing Hua University — **Sept 2019 - June 2024**
Bachelor of Science in Computer Science; GPA: 3.91/4.0 — Hsinchu, Taiwan

Skills

Programming Languages: Python, Java, C/C++, JavaScript, TypeScript, Kotlin, HTML/CSS
Frameworks: Spring Boot, React
Developer Tools: Git, Azure, GCP, RESTful API, Swagger UI, Cypress, Selenium, Android Studio, Figma, Firebase
Databases & Cloud/Infra: MySQL, MongoDB, Prometheus
ML Tools: Pytorch, Tensorflow, NLP, LLM, OpenCV, Keras, Numpy, Pandas, Matplotlib, MATLAB

Work Experience

TSMC - AI Application Integration Division — **June - Aug 2026**
Software Engineer Intern | Java, Spring Boot — Hsinchu, Taiwan

- Engineered the backend of an Agentic SRE monitoring platform, structuring a Controller-Service-DAO architecture to integrate MongoDB, Prometheus, and internal AI prediction APIs into a unified system
- Built dynamic monitoring parameters and an intelligent alerting module to support **10+** fabs, reducing Mean Time To Mitigate (MTTM) by an estimated **10%**
- Adopted Spec-Kit for spec-driven development to accelerate the iteration process, reducing requirement-to-implementation planning time by **50%**

CloudMile - Product Team — **Mar - July 2025**
Software Engineer Intern | Python, Cypress — Taipei, Taiwan

- Authored **800+** detailed test cases for **2** client-facing SaaS products, ensuring **100%** coverage of core user flows, edge cases, and UI/UX behavior based on PRDs and design mockups
- Increased test coverage to **90%** and reduced manual QA time by **70%** by introducing Cypress for Lumiture.ai
- Prioritized testing for **1200+** companies by collaborating with **5+** cross-functional roles to clarify requirements

Garmin - Garmin Golf App (GGA) Team — **July 2023 - May 2024**
Software Engineer Intern | Kotlin, Java, Jetpack Compose — New Taipei, Taiwan

- Developed highly interactive UI components for the GGA 3.0 major release using Jetpack Compose, including real-time scoring leaderboards and dynamic gaming cards for **1M+** active users.
- Modernized UI architecture by refactoring **10+** core functions into reusable components, achieving a **13%** reduction in code redundancy and significantly enhancing code maintainability and scalability.
- Collaborated within an agile international team of 10+ engineers, ensuring high-quality delivery through **20+** impactful code reviews and cross-functional technical alignment.

Information Security Lab — **Sept 2022 - Dec 2023**
Undergraduate Researcher | Python — Hsinchu, Taiwan

- Ranked in the top **20** teams in the Taiwan Futures Exchange Mock Competition by developing a Bollinger Bands strategy with custom calculations and ATR-based stop-loss/take-profit for risk management
- Achieved **45%** prediction accuracy by optimizing market trend prediction with Keras, adapting data preprocessing from 30-minute to 1-minute K-line intervals for timely signal execution

Selected Projects

Crowd Energy Detection — **Oct - Dec 2025**
Python

- Built an end-to-end automated data pipeline integrating the Google Sheets API for dual-annotator workflows, automating video streaming and clipping via yt-dlp and ffmpeg
- Engineered multi-modal feature extraction (Audio RMS, Optical Flow, Pose) to generate structured .npz datasets, significantly reducing manual data preparation time from hours to minutes
- Integrated Scikit-learn ML models (Random Forest, Ridge Regression) into the pipeline as a unified evaluation component, achieving MAE of **5.74s** and outperforming the rule-based baseline by **62.5%**